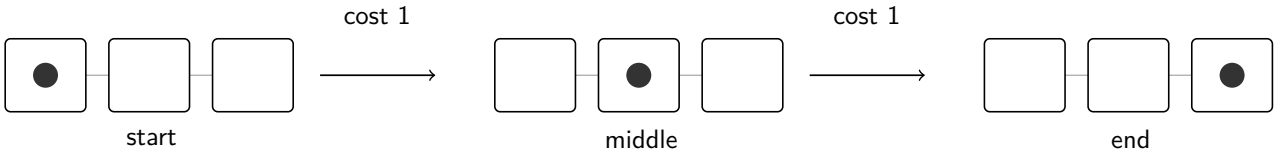
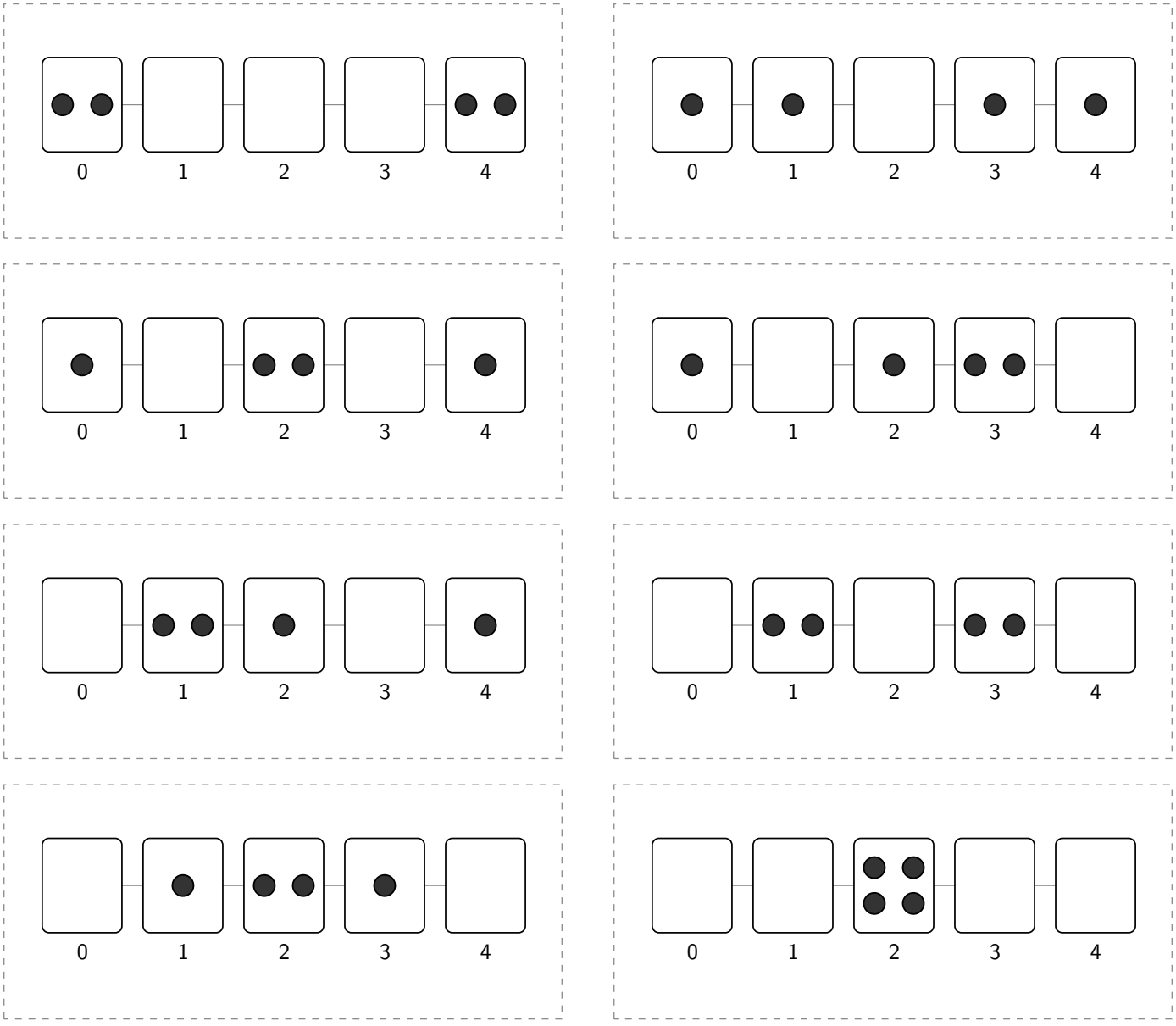
Move one counter at a time to a neighboring bin. Each crossing costs 1. Bins hold any number of counters, and counters may pass through occupied bins. Any counter may fill any target place. Only start counters move; the target picture stays fixed. Write each move as you make it.



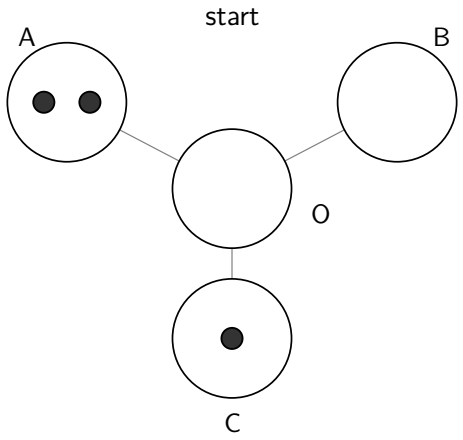
One counter in bin 1 and one in bin 3 have position total $1 + 3 = 4$.

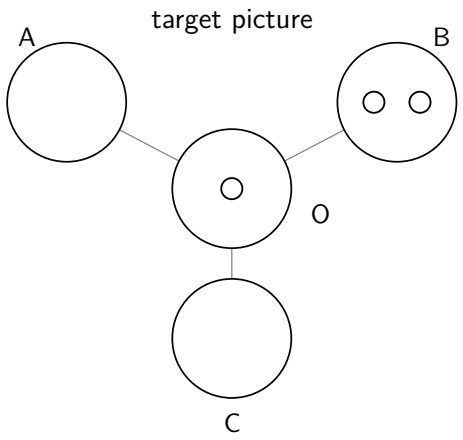
Problem 4: Each card has four counters with position total 8. Which two need the most work to change one into the other? Explain why no pair needs more. Is any four-counter arrangement on bins 0 to 4 with position total 8 missing?



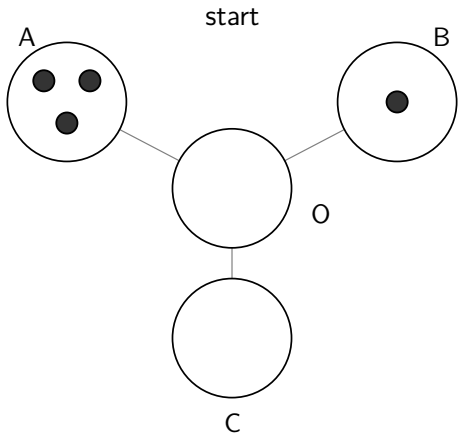
work:

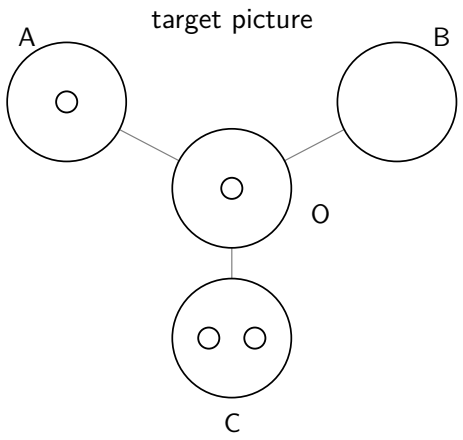
Problem 5: Move counters along the roads. One counter crossing one road costs 1. Find the least work for both pairs. Find a way to know the least work before moving. Does it work for every start and target with the same number of counters on these roads?



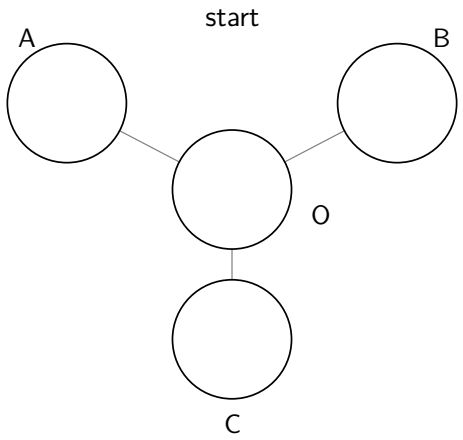


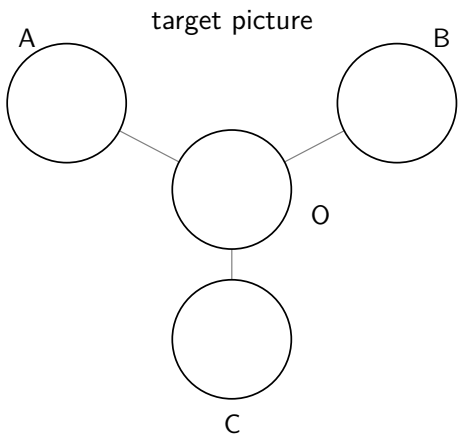
work: _____





work: _____





Problem 6: Make three arrangements, A, B, and C, with four counters in each. Can the least work from A to B plus the least work from B to C be less than the least work straight from A to C? Explain. Find examples where the amounts are equal and where they are different.

